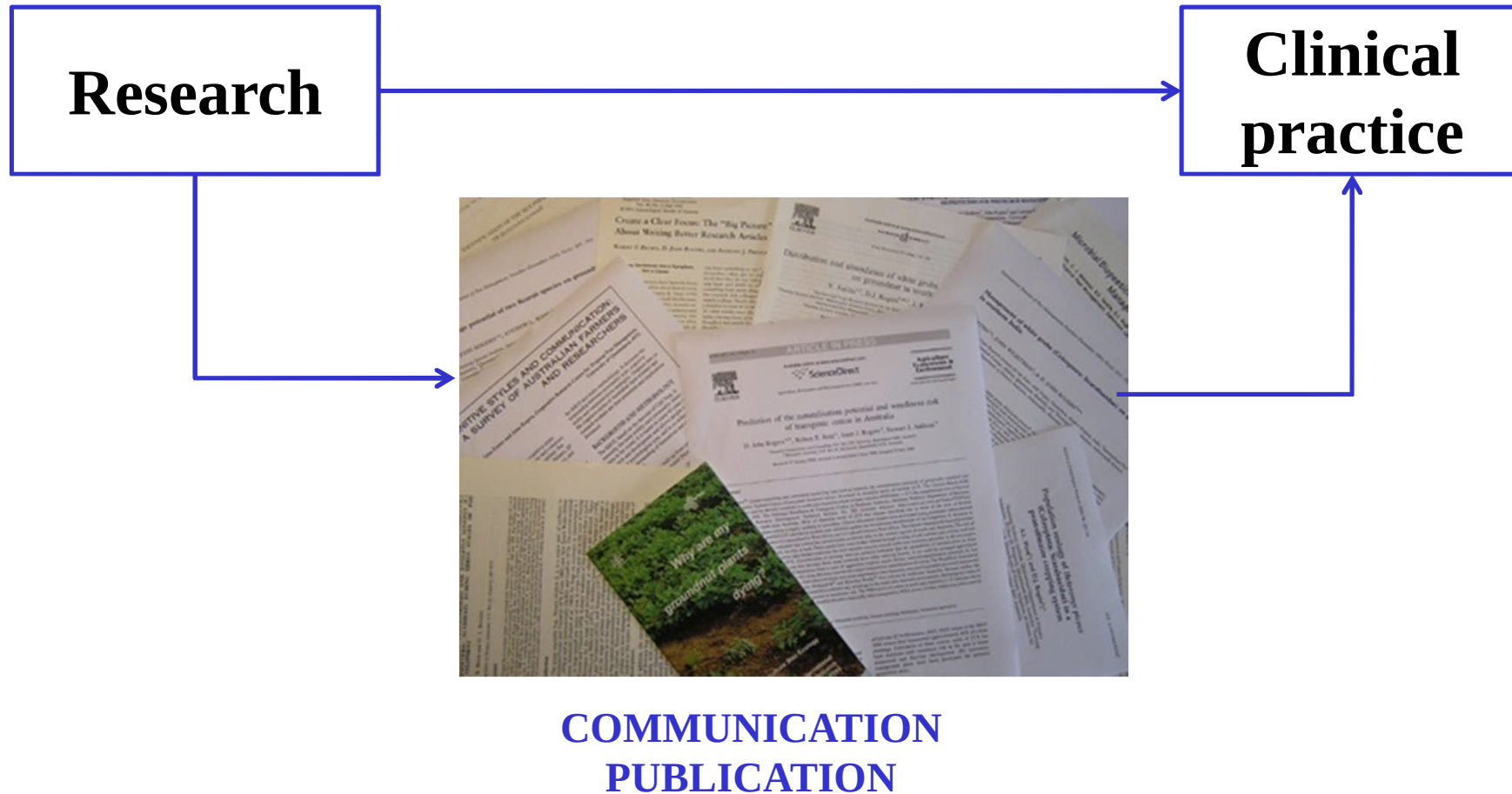


Spin in research publications

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Transposition of research results



Interpretation of trials' results

- In RCTs, data should speak for themselves¹
- Scientists and sponsors are rarely neutral regarding the results of their trial
 - Delay and level of publication
 - Use of the experimental treatment in clinical practice
 - Future career and profit
- Broad latitude in writing an article

➔ Bias the presentation and interpretation of the results²⁻⁴



Kaptchuk T, BMJ, 2003

¹ Fletcher, Med Law, 2007

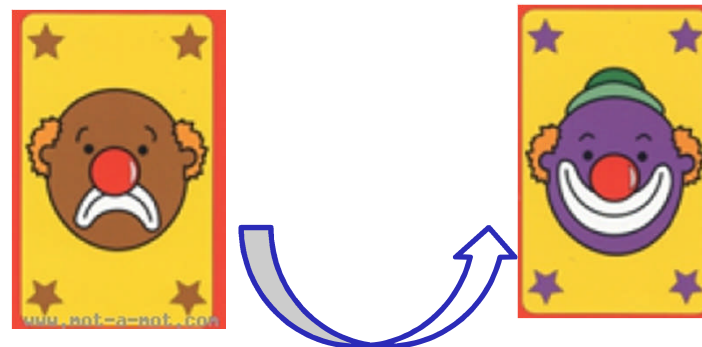
² Horton, BMJ, 1995

³ Yank V, BMJ, 2007

⁴ Jefferson T, BMJ, 2009

Spin in published RCTs

- “Spin” as a way of reporting to convince the reader that
 - the beneficial effect (efficacy, safety) of the experimental treatment is higher than shown by the results



RESEARCH METHODOLOGY

Heed the data when results are not significant

Unexpected non-significant results from randomised trials can be difficult to accept.

Catherine Hewitt, Natasha Mitchell, and David Torgerson find that some authors continue to support interventions despite evidence that they might be harmful

- **17** Negative trials published in the BMJ 2002-2006 (p value 0.05-0.3)
- 4/17: Recommendation to use the treatment in the “what this study adds” box

Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

Isabelle Boutron, MD, PhD

Susan Dutton, MSc

Philippe Ravaud, MD, PhD

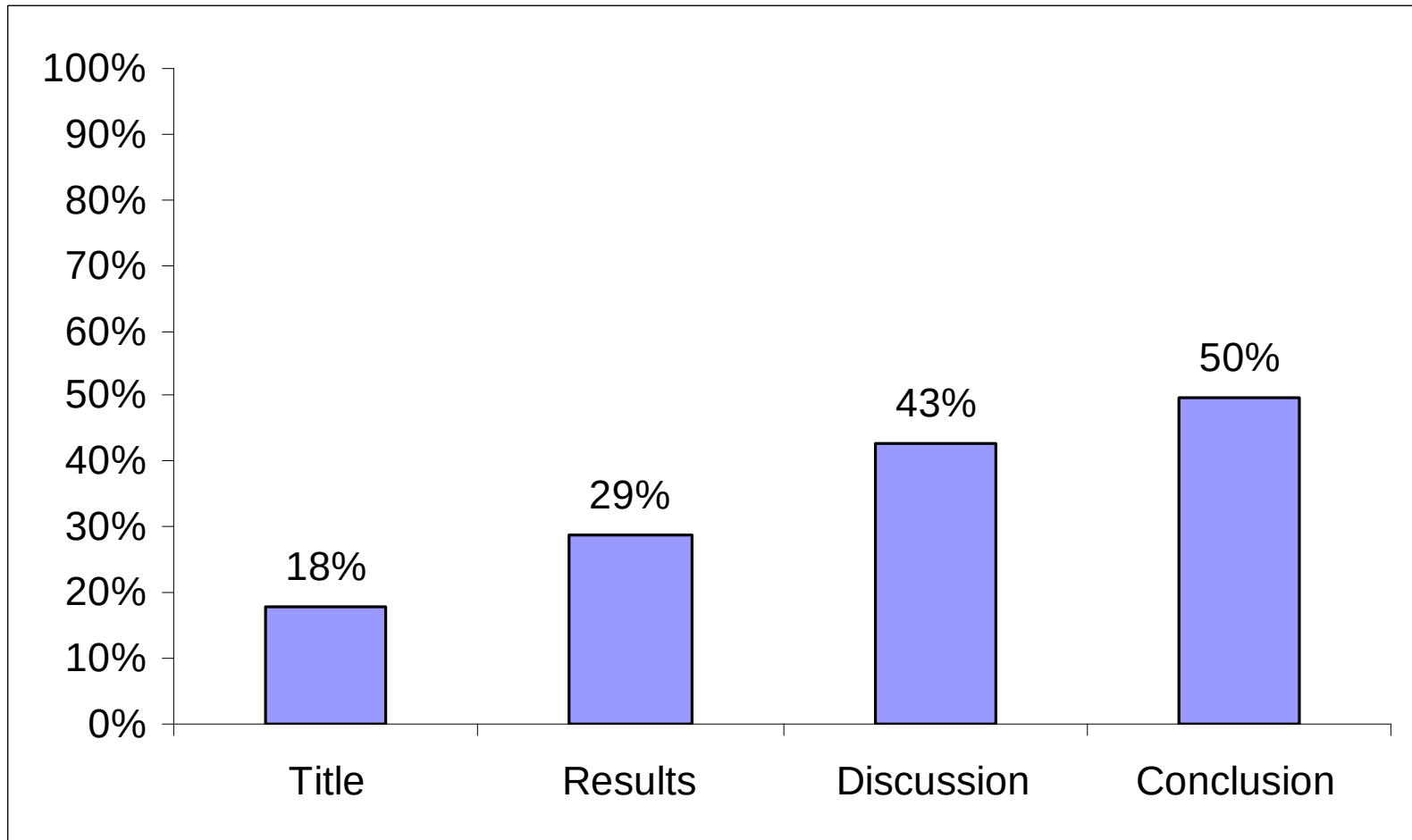
Douglas G. Altman, DSc

Context Previous studies indicate that the interpretation of trial results can be distorted by authors of published reports.

Objective To identify the nature and frequency of distorted presentation or “spin” (ie, specific reporting strategies, whatever their motive, to highlight that the experimental treatment is beneficial, despite a statistically nonsignificant difference for the

- To identify “spin” in a representative sample of 72 negative RCTs (i.e., $p \geq 0.05$).

Prevalence of “spin” in the article



>40% of the reports had “spin” in at least 2 of the 3 sections of the main text

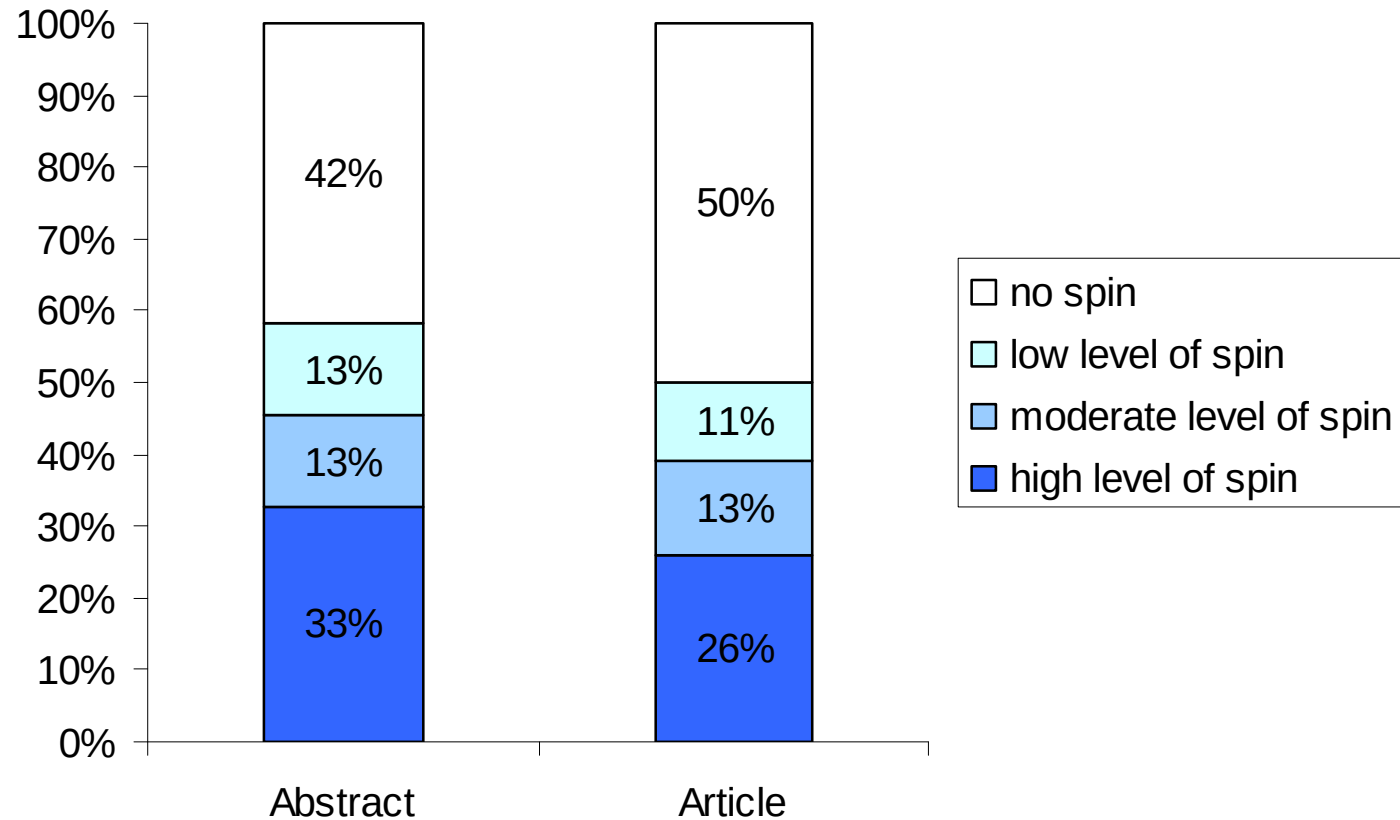
Categories of “spin” in the conclusion

	N=72
Claim of equivalence or comparable effectiveness	6 (8%)
Particular focus on statistically significant results	8 (11%)
Linguistic “spin” only	5 (7%)
[The treatment] “ <i>is expected to be a very important modality in the treatment strategy</i> ”	
[The treatment effect] “ approached , but did not achieve conventional statistical significance”	

Categories of spin in the conclusion

Other spin in the conclusion	22 (31 %)
• Ruling out an adverse event E.g., [...] "we have demonstrated (for the first time) that [...] with the treatment...], embryo implantation is unaltered "	3 (4%)
• Focusing on overall group assessment E.g., "... the mean improvement [...] was clinically relevant in both treatment groups"	9 (13%)
• Recommendation to use the treatment	3 (4%)
• Focus on another objective E.g., reporting the results as if it was not a comparative RCT	3 (4%)
• Other E.g., "Both [..treatments..] were shown to be superior to a historical placebo control"	4 (6%)

Level of spin in the conclusion



High level of “spin” : no acknowledgement of the negative primary outcome
no uncertainty,
no recommendation for further trial

RCTs results dissemination

Randomized controlled trials



Scientific publication, congress
communication

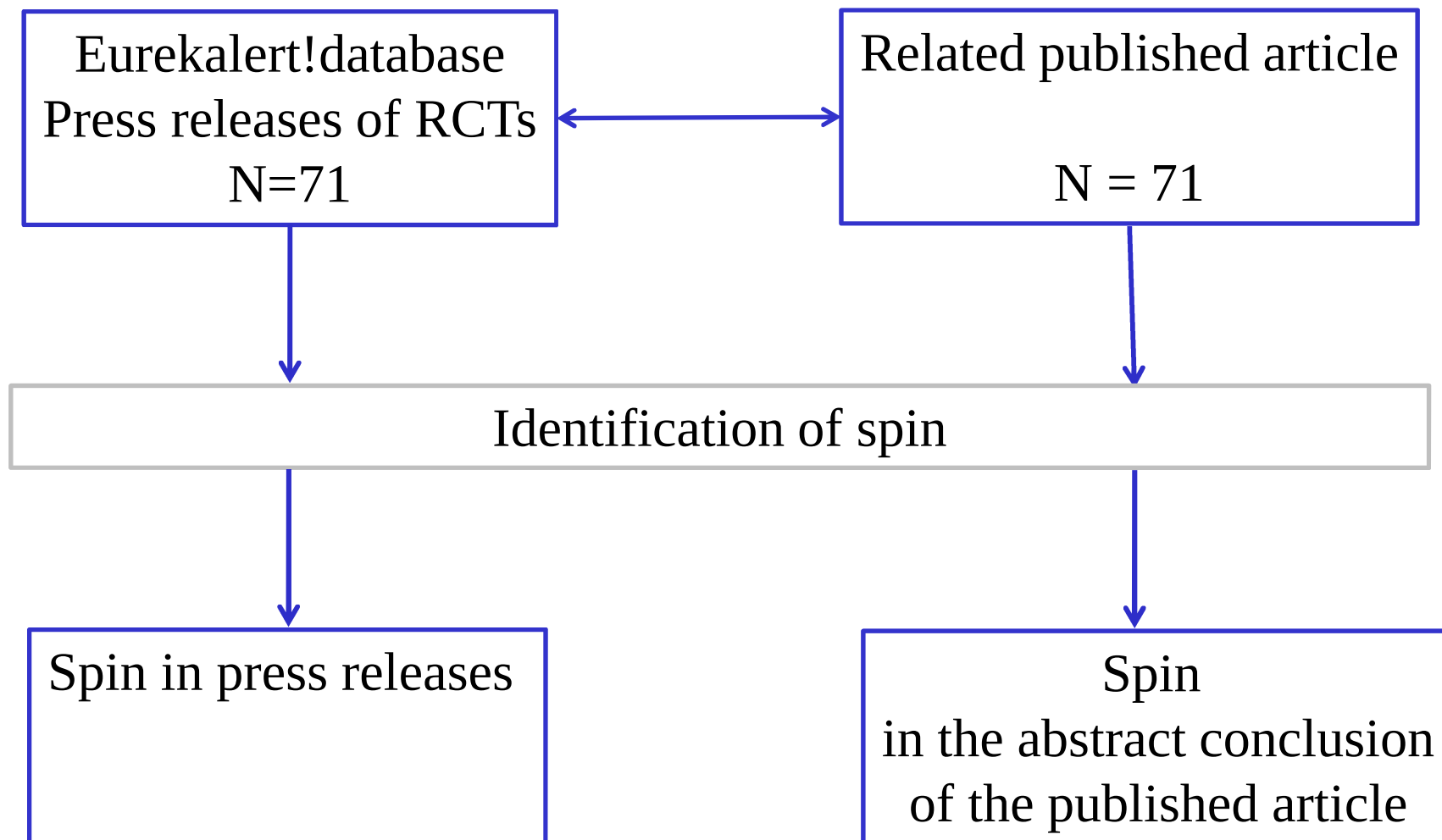


Press releases



Lay press

Spin in press releases



Spin in press releases

34 (47%) No spin in the published abstract conclusion (Ccl) and the PR

8 (11%) No spin in the abstract Ccl but spin added in the PR

4 (6%) Spin in the abstract Ccl deleted in the PR

12 (17%) Same spin in the abstract Ccl and the PR

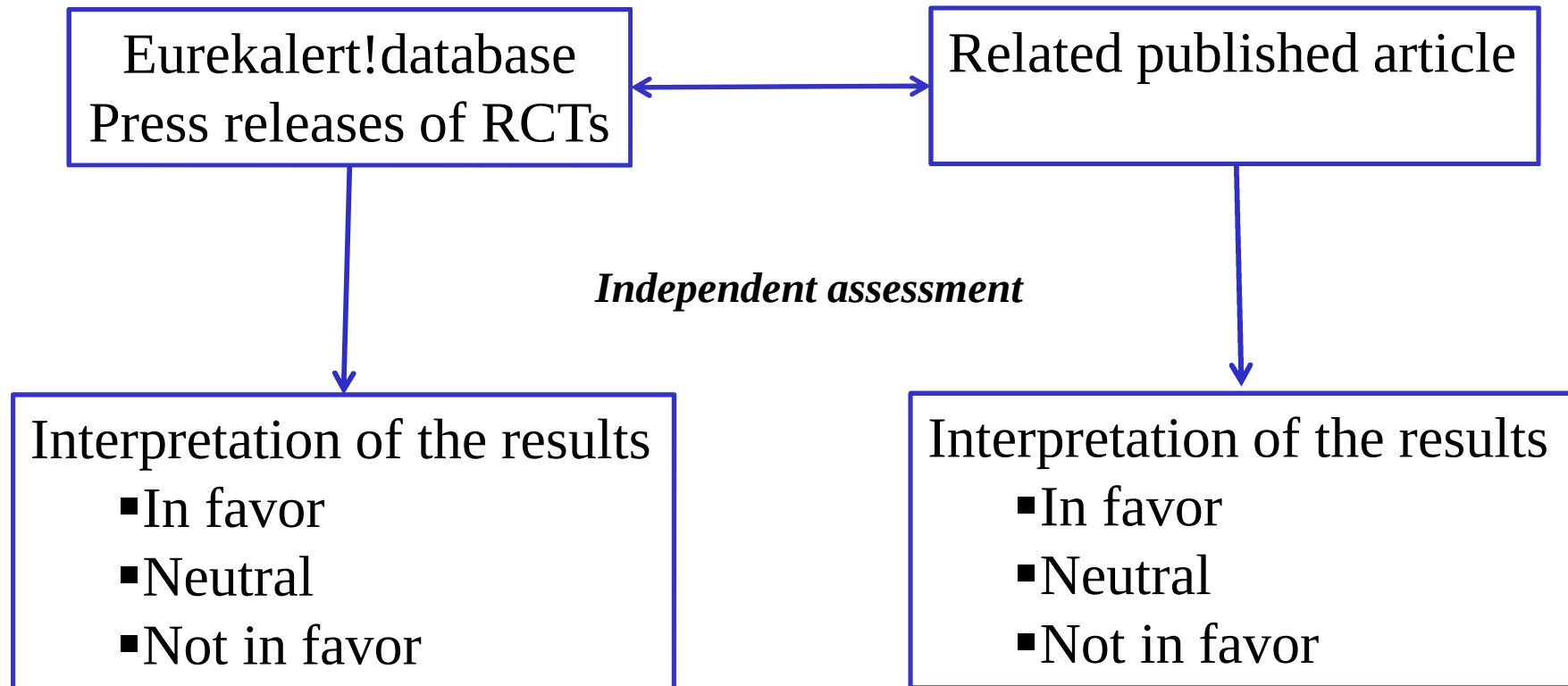
13 (18%) Same spin in the abstract Ccl and the PR and new spin added

Factors associated with spin in PR¹:

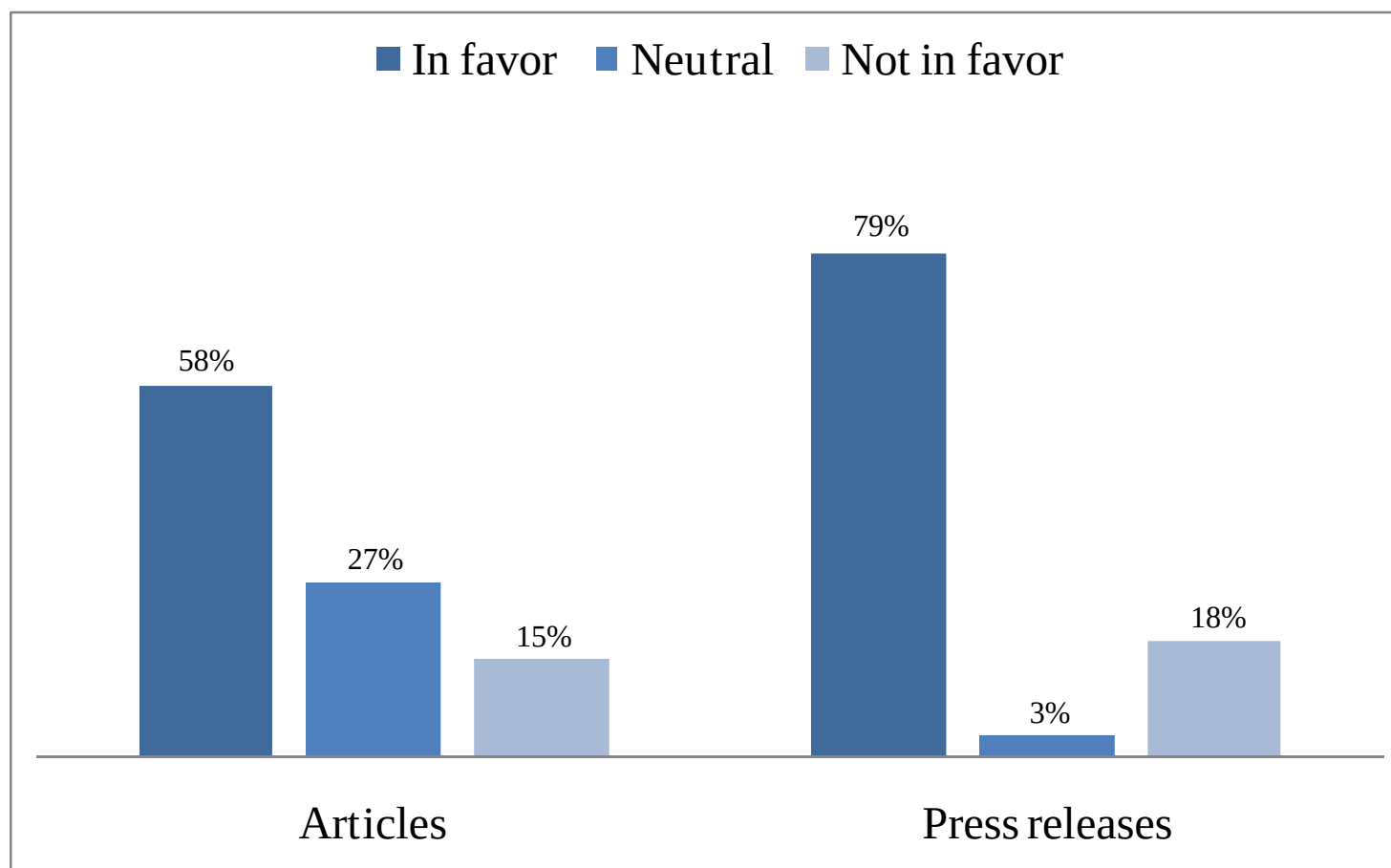
Spin in RCT published abstract conclusion: RR = **3.8 [1.5-9.3]**

Multivariate analysis: sample size, impact factor, spin in abstract conclusion, journal, results of the primary outcome

Spin in press releases



Spin and press releases



Inadequate interpretation from press releases: **22/71 CP (31%)**

- **19/22 (86%):** overestimation of treatment beneficial effect
- **3/22 (14%):** underrestimation of treatment beneficial effect

Spin and press releases

- Factors associated with overestimation of treatment beneficial effect

	yes	no	
Specialist journal	45%	6%	p=0.002
Negative primary outcome	42%	19%	p=0.05
Sample size (< 110)	46%	8%;	p=0.0004
Spin in the press release	55%	3%	p=0.0004

*Yavchitz A, Boutron I, Bafeta A, Marroun I, Charles P, Mantz J, Ravaud P.
Press Releases could induce misinterpretation of research results. **submitted***

Conclusions

- “Spin” in reports of RCTs are not rare
- More research in this field is needed